CIS 240 LAB REPORT

Switching, Routing, Wireless Essentials / Enterprise Networks, Security, and Automation

2.2.13 Packet Tracer — Point-to-Point Single-Area OSPFv2 Configuration

BACKGROUND	
Course:	CIS 240
Term:	Spring 2026
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Submission Platform:	Canvas / Cisco NetAcad / Packet Tracer Activity Submission as Assigned
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Report Title:	2.2.13 Packet Tracer - Point-to-Point Single-Area OSPFv2 Configuration
Lab Type:	Hands-On Lab / Packet Tracer
Date Completed:	2026-04-22
Time Spent:	Approximately 1.0 hour

ENVIRONMENT

Platforms, Devices, and Systems Used:

- Cisco Packet Tracer simulation environment used for router configuration, protocol verification, and final validation.
- Routing devices used: R1, R2, and R3.
- End systems used: PC1, PC2, and PC3, with one LAN attached to each router.
- Access switches used: one switch per LAN segment as shown in the activity topology.
- WAN design used: triangular point-to-point serial topology interconnecting all three routers.
- Routing protocol used: OSPFv2 in a single-area deployment using Area 0.
- Configuration focus: explicit router IDs, wildcard masks, quad-zero wildcard masks, interface-level OSPF configuration, passive interfaces, adjacency formation, and routing-table verification.

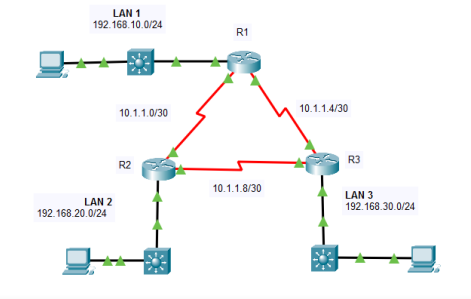

Figure 1. Initial topology showing the three-router triangular WAN design with one LAN attached to each router before final verification.

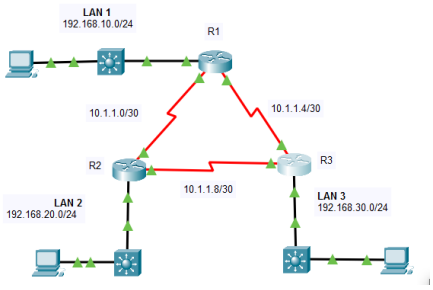

Figure 2. Final topology after OSPFv2 configuration, passive-interface setup, and full routing convergence across the routed network.

Topology Overview:

- This activity used a three-router topology in which each router had one LAN-facing GigabitEthernet interface and two serial WAN links.
- The serial interfaces formed a triangle so that every router had two OSPF neighbors once the lab was complete.
- Each router also advertised one /24 LAN into the OSPF domain.
- The activity was especially useful because it required three different OSPF configuration styles in the same network while still converging into one single-area routing domain.

Addressing Plan Used:

- R1 G0/0/0: 192.168.10.1 /24
- R1 S0/1/0: 10.1.1.1 /30
- R1 S0/1/1: 10.1.1.5 /30
- R2 G0/0/0: 192.168.20.1 /24
- R2 S0/1/0: 10.1.1.2 /30
- R2 S0/1/1: 10.1.1.9 /30
- R3 G0/0/0: 192.168.30.1 /24
- R3 S0/1/0: 10.1.1.10 /30
- R3 S0/1/1: 10.1.1.6 /30
- PC1: 192.168.10.10 /24
- PC2: 192.168.20.10 /24
- PC3: 192.168.30.10 /24
- OSPF process ID used on all routers: 10
- OSPF area used on all participating interfaces: 0

Work Objective:

- Configure OSPFv2 across all three routers in a single-area topology.
- Assign explicit router IDs to ensure deterministic OSPF identification and predictable verification output.
- Configure R1 using subnet-based network statements and wildcard masks.
- Configure R2 using interface IP addresses with quad-zero wildcard masks.
- Configure R3 using interface-level OSPF activation on the required interfaces.
- Configure LAN-facing interfaces as passive and verify full route convergence across the topology.

PART 1 — OSPF PROCESS AND ROUTER IDS

Router ID Configuration Procedure:

1) Opened the CLI on R1, R2, and R3 and entered privileged EXEC mode with enable.
2) Entered global configuration mode with configure terminal.
3) Started the OSPF routing process on each router with router ospf 10.
4) Assigned explicit router IDs rather than relying on automatic highest-interface-address selection.
5) Configured R1 with router-id 1.1.1.1.
6) Configured R2 with router-id 2.2.2.2.
7) Configured R3 with router-id 3.3.3.3.
8) Kept the process ID consistent at 10 on all routers so the activity matched the required design exactly.
9) Verified mentally before proceeding that the router IDs were unique and that no duplicate OSPF identifiers existed in the topology.

Commands Applied During Part 1:

BASE ACCESS	ROUTER IDS	PURPOSE
enable	router-id 1.1.1.1	OSPF process 10
configure terminal	router-id 2.2.2.2	unique router IDs
router ospf 10	router-id 3.3.3.3	deterministic verification

Operational Notes:

- Router IDs uniquely identify OSPF routers inside the routing domain.
- Manual router IDs are preferred because they remain stable even if interface addressing changes later.
- This part established the base OSPF process before any networks were advertised or any interfaces were enabled for OSPF participation.
- Using explicit router IDs also made the verification output easier to interpret during show ip protocols and routing validation.
- The assigned router IDs matched the activity exactly, which mattered because the Packet Tracer assessment checked the values literally.

Part 1 Result:

- Part 1 was completed successfully because OSPF process 10 was created on all routers and the router IDs matched the required values of 1.1.1.1, 2.2.2.2, and 3.3.3.3.

PART 2 — OSPF NETWORK CONFIGURATION METHODS

R1 Network-Statement Method:

- R1 was configured using subnet-based network statements.
- The LAN network 192.168.10.0/24 required wildcard mask 0.0.0.255.
- The two serial /30 networks required wildcard mask 0.0.0.3.
- Under router ospf 10, the following commands were applied:
  network 192.168.10.0 0.0.0.255 area 0
  network 10.1.1.0 0.0.0.3 area 0
  network 10.1.1.4 0.0.0.3 area 0
- This method matched full subnets rather than exact interface IP addresses and reflected the classic CCNA-style OSPF configuration approach.

R2 Quad-Zero Method:

- R2 was configured using interface IP addresses combined with the quad-zero wildcard mask 0.0.0.0.
- This method matched only the exact interface address that was entered.
- Under router ospf 10, the following commands were applied:
  network 192.168.20.1 0.0.0.0 area 0
  network 10.1.1.2 0.0.0.0 area 0
  network 10.1.1.9 0.0.0.0 area 0
- This method was more precise than the subnet-based approach and demonstrated that OSPF can be targeted directly to exact interface addresses.

PART 2 — OSPF NETWORK CONFIGURATION METHODS (CONTINUED)

R3 Interface-Level Method:

- R3 was configured by enabling OSPF directly on participating interfaces instead of using network statements.
- This method is interface-specific and avoids wildcard-mask calculations entirely.
- The following commands were entered:

```
interface g0/0/0
ip ospf 10 area 0
interface s0/1/0
ip ospf 10 area 0
interface s0/1/1
ip ospf 10 area 0
```

- This approach made the active OSPF interfaces explicit and aligned exactly with the activity requirement for router R3.

Passive Interface Configuration:

- LAN-facing interfaces were configured as passive so OSPF would advertise the connected LAN networks without sending hello packets toward end hosts.
- The following command was applied on all routers under router ospf 10:

```
passive-interface g0/0/0
```

- This prevented unnecessary routing-protocol traffic from leaving the user-facing GigabitEthernet interface while still allowing the connected subnet to appear in the OSPF database and routing table.

Configuration Summary by Router:

- R1 used classic subnet network statements with calculated wildcard masks.
- R2 used host-address network statements with quad-zero masks for exact interface matching.
- R3 used direct interface activation with ip ospf 10 area 0.
- All three routers used passive-interface g0/0/0 for the LAN-facing interface.
- Together, the three methods demonstrated that OSPF can be configured flexibly while still producing the same converged routing outcome.

Full Command Set Used in the Lab:

R1

```
enable
configure terminal
router ospf 10
router-id 1.1.1.1
network 192.168.10.0 0.0.0.255
area 0
network 10.1.1.0 0.0.0.3 area 0
network 10.1.1.4 0.0.0.3 area 0
passive-interface g0/0/0
end
write memory
```

R2

```
enable
configure terminal
router ospf 10
router-id 2.2.2.2
network 192.168.20.1 0.0.0.0 area
0
network 10.1.1.2 0.0.0.0 area 0
network 10.1.1.9 0.0.0.0 area 0
passive-interface g0/0/0
end
write memory
```

R3

```
enable
configure terminal
router ospf 10
router-id 3.3.3.3
passive-interface g0/0/0
exit
interface g0/0/0
ip ospf 10 area 0
exit
interface s0/1/0
ip ospf 10 area 0
exit
interface s0/1/1
ip ospf 10 area 0
end
write memory
```

Part 2 Result:

- Part 2 was completed successfully because all required networks were placed into OSPF using the correct method for each router and all LAN interfaces were configured as passive.

PART 3 — VERIFICATION AND ROUTING VALIDATION

Verification Commands Used:

- show ip protocols
- show ip ospf interface brief
- show ip route
- show running-config | section ospf

Verification Interpretation:

- show ip protocols confirmed the OSPF process number, router ID, advertised networks, and passive-interface settings.
- show ip ospf interface brief confirmed which interfaces were participating in OSPF and verified that all active interfaces were placed in Area 0.
- show ip route confirmed that remote LAN networks were learned dynamically as OSPF routes marked with O.
- show running-config | section ospf confirmed that the final running configuration matched the intended design for each router.
- The combined outputs demonstrated adjacency formation across the serial links, correct route installation, and complete route exchange between all three routed LANs.

```
R1#show ip protocols

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.10.0 0.0.0.255 area 0
    10.1.1.0 0.0.0.3 area 0
    10.1.1.4 0.0.0.3 area 0
  Passive Interface(s):
    GigabitEthernet0/0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:18:33
    2.2.2.2              110      00:19:06
    3.3.3.3              110      00:25:42
  Distance: (default is 110)

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/1/0
L       10.1.1.1/32 is directly connected, Serial0/1/0
C       10.1.1.4/30 is directly connected, Serial0/1/1
L       10.1.1.5/32 is directly connected, Serial0/1/1
O       10.1.1.8/30 [110/128] via 10.1.1.2, 00:18:41, Serial0/1/0
                    [110/128] via 10.1.1.6, 00:18:41, Serial0/1/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
O    192.168.20.0/24 [110/65] via 10.1.1.2, 00:28:57, Serial0/1/0
O    192.168.30.0/24 [110/65] via 10.1.1.6, 00:18:41, Serial0/1/1

R1#show running-config | section ospf
router ospf 10
 router-id 1.1.1.1
 log-adjacency-changes
 passive-interface GigabitEthernet0/0/0
 network 192.168.10.0 0.0.0.255 area 0
 network 10.1.1.0 0.0.0.3 area 0
 network 10.1.1.4 0.0.0.3 area 0
R1#
```

Figure 3. R1 verification showing configured networks, OSPF process information, and learned remote routes.

```
R2#show ip protocols

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 2.2.2.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.20.1 0.0.0.0 area 0
    10.1.1.2 0.0.0.0 area 0
    10.1.1.9 0.0.0.0 area 0
  Passive Interface(s):
    GigabitEthernet0/0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:19:16
    2.2.2.2              110      00:19:49
    3.3.3.3              110      00:26:25
  Distance: (default is 110)

R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/1/0
L       10.1.1.2/32 is directly connected, Serial0/1/0
O       10.1.1.4/30 [110/128] via 10.1.1.1, 00:19:56, Serial0/1/0
                    [110/128] via 10.1.1.10, 00:19:56, Serial0/1/1
C       10.1.1.8/30 is directly connected, Serial0/1/1
L       10.1.1.9/32 is directly connected, Serial0/1/1
O    192.168.10.0/24 [110/65] via 10.1.1.1, 00:29:40, Serial0/1/0
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/0
O    192.168.30.0/24 [110/65] via 10.1.1.10, 00:19:56, Serial0/1/1

R2#show running-config | section ospf
router ospf 10
 router-id 2.2.2.2
 log-adjacency-changes
 passive-interface GigabitEthernet0/0/0
 network 192.168.20.1 0.0.0.0 area 0
 network 10.1.1.2 0.0.0.0 area 0
 network 10.1.1.9 0.0.0.0 area 0
R2#
```

Figure 4. R2 verification showing quad-zero network statements and final route propagation.

```
R3>enable
R3#show ip protocols

Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 3.3.3.3
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
  Passive Interface(s):
    GigabitEthernet0/0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:20:05
    2.2.2.2              110      00:20:38
    3.3.3.3              110      00:27:14
  Distance: (default is 110)

R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O       10.1.1.0/30 [110/128] via 10.1.1.9, 00:20:13, Serial0/1/0
                    [110/128] via 10.1.1.5, 00:20:13, Serial0/1/1
C       10.1.1.4/30 is directly connected, Serial0/1/1
L       10.1.1.6/32 is directly connected, Serial0/1/1
C       10.1.1.8/30 is directly connected, Serial0/1/0
L       10.1.1.10/32 is directly connected, Serial0/1/0
O    192.168.10.0/24 [110/65] via 10.1.1.5, 00:20:13, Serial0/1/1
O    192.168.20.0/24 [110/65] via 10.1.1.9, 00:20:44, Serial0/1/0
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0/0

R3#show running-config | section ospf
 ip ospf 10 area 0
 ip ospf 10 area 0
 ip ospf 10 area 0
router ospf 10
 router-id 3.3.3.3
 log-adjacency-changes
 passive-interface GigabitEthernet0/0/0
R3#
```

Figure 5. R3 verification showing interface-based OSPF configuration and final routing convergence.

What the Verification Proved:

- OSPF process 10 was active on all routers.
- Router IDs were correct and stable.
- R1 advertised networks with wildcard-based subnet statements.
- R2 advertised exact interface IP addresses with quad-zero masks.
- R3 participated in OSPF through direct interface activation.
- G0/0/0 was passive on all three routers.
- Each router learned the remote /24 LANs through OSPF.
- The serial links formed the neighbor relationships required for route exchange.

Part 3 Result:

- Part 3 was completed successfully because all routers formed adjacencies across the serial links, all remote LAN networks were learned through OSPF, and the verification output confirmed the expected final state.

SUMMARY, RESULTS, AND REFLECTION

Completion Verification:

- OSPF process 10 was configured on R1, R2, and R3.
- Router IDs were set correctly as 1.1.1.1, 2.2.2.2, and 3.3.3.3.
- R1 used subnet network statements with wildcard masks.
- R2 used interface IP addresses with quad-zero wildcard masks.
- R3 used interface-based OSPF configuration.
- G0/0/0 was configured as a passive interface on all routers.
- OSPF adjacencies formed correctly across the serial links joining the three routers.
- Routing tables showed dynamic OSPF-learned routes for all remote LAN segments.
- All three LAN networks were successfully advertised into Area 0.
- The final topology operated with full convergence and end-to-end reachability between the routed networks.

Key Findings:

1) OSPF can be deployed using multiple configuration styles while still producing the same routing outcome.
2) Wildcard-mask calculations remain essential when using traditional network statements.
3) Quad-zero matching is precise and useful when exact interface selection is desired.
4) Interface-level OSPF configuration is clean and direct when the design requires only selected interfaces to participate.
5) Passive interfaces are important for reducing unnecessary routing protocol traffic on user-facing LAN segments.
6) Verification commands such as show ip protocols, show ip ospf interface brief, and show ip route are most effective when used together.

Reflection:

- This lab reinforced both the flexibility and structure of OSPFv2.
- The most valuable takeaway was seeing the same routing protocol configured three different ways in one topology while still producing a fully converged network.
- The completed work demonstrates the ability to interpret an addressing table, apply router-specific OSPF methods, configure passive interfaces correctly, and validate final operation using standard Cisco IOS verification commands.
- From an operational standpoint, the lab also reinforced that correct verification is just as important as the initial configuration itself.

Deliverables:

- Completed Packet Tracer file for 2.2.13 Point-to-Point Single-Area OSPFv2 Configuration.
- Final written lab report documenting topology, configuration workflow, verification, and outcomes.
- Initial and final topology views integrated into the report.
- Router verification views for R1, R2, and R3 integrated into the report.